

EV Battery Charging Using CC-CV Method

Simulation Study in MATLAB/Simulink

1. Introduction

Charging lithium-based cells in a reliable and efficient manner is a fundamental requirement for any battery-powered electric vehicle (EV). Among the various strategies available, the Constant Current – Constant Voltage (CC-CV) technique has established itself as the industry standard owing to its simplicity, effectiveness, and compatibility with a wide range of cell chemistries.

This report documents a simulation of the CC-CV charging process applied to a Lithium-Ion battery cell using MATLAB/Simulink. The simulation tracks key parameters such as State of Charge (SOC), terminal voltage, and charging current across both phases of the charging cycle.

2. Battery Specifications

The battery cell used in this simulation has the following characteristics:

Parameter	Value
Cell Chemistry	Lithium Iron Phosphate (LFP)
Nominal Voltage	3.3 V
Rated Capacity	2.3 Ah
Initial State of Charge (SOC)	0%
CC Phase Cutoff SOC	80%
Charging Cutoff SOC	90%
CC Phase Current (cu)	2A
CV Phase Voltage	Equal to nominal cell voltage (3.3 V)

3. CC-CV Charging Method – Overview

The CC-CV method divides battery charging into two distinct phases. In practice, starting a charge from a very low SOC and applying a well-defined transition point between the two phases protects cell integrity and maximises the usable energy stored.

3.1 Constant Current (CC) Phase

During this phase the charger delivers a fixed magnitude of current into the cell regardless of the instantaneous terminal voltage. The cell voltage climbs steadily as charge is accumulated. In this simulation the CC phase runs from an initial SOC of 0% and continues until the SOC reaches 80%. The switch S1 in the Simulink model remains active throughout this interval, while S2 remains open.

3.2 Constant Voltage (CV) Phase

Once the SOC crosses the 80% threshold, the model transitions to the CV phase. The terminal voltage is now held at the nominal cell voltage (3.3 V) and the current is allowed to decay naturally as the cell approaches full charge. The phase terminates when the SOC reaches 90%, which is the defined charging cutoff for this study. Limiting the maximum SOC to 90% rather than 100% is a deliberate design choice to reduce lithium plating risk and prolong cycle life.

4. Simulink Model Description

The Simulink schematic implements the CC-CV strategy using a set of functional blocks arranged into two parallel control paths, one for each charging phase. The overall solver is configured in discrete mode with a fixed step size of 5e-05 s using the powergui block.

4.1 Battery Block

A generic battery block is placed at the centre of the model. It exposes three measurement outputs – terminal voltage (V), charging current (A), and state of charge (SOC %). These signals are routed to the powergui scope for monitoring and also fed back into the control logic.

4.2 SOC-Based Logic Block (fcn)

A MATLAB Function block labelled fcn receives the real-time SOC value as its input. Based on the current SOC, it computes three outputs:

- s1 – gate signal for switch S1 (active during CC phase, SOC < 80%)
- s2 – gate signal for switch S2 (active during CV phase, SOC ≥ 80%)
- cu – the reference current magnitude passed to the CC path comparator

4.3 CC Path (Upper Subsystem)

The CC path contains a comparator that receives the reference current cu and the measured battery current. The error signal drives a pulse-width modulated gate signal for switch S1. As long as SOC is below 80%, S1 conducts and the battery is supplied with constant current. The output of this subsystem is connected to a display block labelled CC showing the active state (0 when inactive).

4.4 CV Path (Lower Subsystem)

The CV path consists of a diode element, a resistor R, and switch S2. When the SOC crosses 80%, the logic block activates s2 and the voltage across the battery settles at 3.3 V.

The resistor limits the current to prevent overshoot at the moment of transition. A display block labelled CV shows the active state (1 when the CV phase is running).

4.5 SOC Display

A constant block shows the real-time SOC value (displayed as 89.89 in the captured screenshot), confirming that the simulation was stopped just before the 90% cutoff. This indicates the model behaved correctly and terminated charging within the defined window.

5. Working Principle of the Simulation

At the start of the simulation the battery SOC is 0%. The fcn logic block sets s1 high and s2 low, enabling the CC path. Current flows through switch S1 at the computed reference magnitude i_{cu} , steadily raising both the terminal voltage and the SOC.

When the SOC crosses 80%, the fcn block toggles the switch signals. S1 is turned off and S2 is activated. The battery terminal voltage is now regulated at 3.3 V. As the internal electrochemical potential of the cell converges towards the applied voltage, the current naturally falls. The simulation detects the SOC reaching 90% and the charging cycle ends. The final SOC recorded in the display is approximately 89.89%, which matches the expected cutoff behaviour.

6. Key Simulation Settings

- Simulation type: Discrete, fixed step
- Solver step size: 5e-05 seconds
- Battery model: Generic (built-in Simulink battery block)
- Nominal voltage: 3.3 V
- Rated capacity: 2.3 Ah
- Initial SOC: 0%
- CC-to-CV transition SOC: 80%
- Simulation stop condition: SOC reaches 90%

7. Results and Observations

The simulation successfully demonstrated the full CC-CV charging cycle from 0% SOC to approximately 90% SOC. The following observations were recorded:

- The CC phase occupied the larger portion of the charging duration, consistent with the fact that the cell is nearly empty at the start.
- Terminal voltage rose gradually throughout the CC phase, which is the expected behaviour for an LFP cell under constant current.
- The transition from CC to CV was smooth, with no visible spike in voltage or current, indicating that the switch logic and resistor sizing are appropriate.
- During the CV phase the current decayed as the cell approached the cutoff SOC, which is consistent with real-world LFP charging profiles.

- The SOC display reading of 89.89% at the end of simulation confirms that the model stopped charging within the defined 90% limit.
- The CC output display showed 0 and the CV display showed 1 at the end of the run, correctly indicating that the CV phase was the last active phase.

8. Advantages of the CC-CV Method for EV Batteries

The CC-CV strategy offers several practical benefits when applied to EV battery packs:

- Straightforward implementation – the two-phase control logic can be realised with minimal hardware and software.
- Reduced thermal stress – constant current charging avoids the voltage spikes seen in simpler methods, generating less heat inside the cell.
- Extended cycle life – terminating at 90% SOC rather than 100% reduces lithium plating and electrolyte degradation over repeated cycles.
- Wide compatibility – the CC-CV profile is supported by most commercial battery management system (BMS) chips.
- Predictable charging time – the CC phase duration can be estimated in advance from the capacity and current rating, aiding scheduling in fleet EV applications.

9. Conclusion

This simulation study examined the CC-CV charging method applied to a 3.3 V, 2.3 Ah LFP battery cell starting from a completely discharged state. The Simulink model captured both charging phases accurately, with the CC phase active from 0% to 80% SOC and the CV phase active from 80% to the 90% cutoff.

The results confirm that the CC-CV approach provides a stable, well-regulated charging profile that is safe for the cell and practical for EV applications. The simulation can be extended in future work to include temperature modelling, pack-level simulations with multiple cells, and hardware-in-the-loop testing to validate the control logic against a physical battery system.